Chapter 3 — Chemical Bonding

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Topic 1 — Types of bonding and electronegativity

Q.1 Explain ionic, covalent and dative bonding, and how electronegativity difference helps predict bond type.

Atoms form bonds to attain a more stable noble-gas-like electronic arrangement. An ionic bond forms by complete transfer of electrons from a low-ionization-energy atom to a high-electron-affinity atom, as in NaCl. A covalent bond forms by mutual sharing of electrons, as in Cl_2 or HCl. A dative bond forms when the shared pair is donated by only one atom, as in NH_4^+ or H_3O^+.

Electronegativity difference gives an approximate guide to bond type. A very small or zero difference gives a non-polar covalent bond, a moderate difference gives a polar covalent bond, and a large difference favours ionic bonding. A molecule may contain polar bonds but still be non-polar overall if its shape makes the bond dipoles cancel, as in CO_2.

Important Conceptual Short Questions — Topic 1

1. Why does NaCl conduct electricity when molten but not when solid?

In solid NaCl the ions are fixed in a lattice and cannot move. In the molten state the ions become mobile, so they can carry charge.

2. How is a dative bond different from an ordinary covalent bond after formation?

The only difference is the origin of the shared pair: in a dative bond both electrons come from one atom. Once formed, it behaves like an ordinary covalent bond.

3. Why is CO_2 non-polar even though C=O bonds are polar?

CO_2 is linear. The two C=O bond dipoles are equal and opposite, so they cancel and the molecule has no net dipole moment.

4. How does electronegativity difference explain the polarity of HCl?

Chlorine is more electronegative than hydrogen, so it attracts the shared pair more strongly. This gives Cl a partial negative charge and H a partial positive charge.

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Topic 2 — Intermolecular forces, bond energy and bond length

Q.2 Describe intermolecular forces and explain the relationship between bond energy, bond length and bond stability.

Intermolecular forces are weaker attractions between molecules. London dispersion forces arise from instantaneous and induced dipoles and increase with molecular size and polarizability. Permanent dipole-dipole forces occur between polar molecules. Hydrogen bonding is a strong dipole attraction when H is bonded to N, O or F.

Bond energy is the energy required to break one mole of bonds in the gaseous state. Bond length is the distance between bonded nuclei. For the same atoms, a triple bond is shorter and stronger than a double bond, and a double bond is shorter and stronger than a single bond. Greater bond energy usually means greater stability and lower reactivity.

Important Conceptual Short Questions — Topic 2

1. Why does water have a much higher boiling point than H_2S?

Water molecules form strong hydrogen bonds, while H_2S molecules mainly have weaker dipole-dipole and dispersion forces. More energy is needed to separate water molecules.

2. How do London forces explain the increase in boiling points of halogens down the group?

Down the group, molecules become larger and more polarizable. Stronger London forces require more energy to overcome, so boiling points increase.

3. Why is a triple bond shorter than a single bond between the same atoms?

A triple bond has more shared electrons and greater electron density between the nuclei. The stronger attraction pulls the nuclei closer together.

4. How does hydrogen bonding affect the density of ice?

Hydrogen bonds hold water molecules in an open hexagonal lattice in ice. This arrangement has more empty space than liquid water, so ice is less dense.

Topic 3 — Valence bond theory and hybridization

Q.3 Explain valence bond theory, sigma and pi bonds, and sp, sp^2 and sp^3 hybridization with examples.

Valence bond theory explains covalent bonding by the overlap of half-filled atomic orbitals. A sigma bond forms by head-on overlap along the internuclear axis and is strong. A pi bond forms by sideways overlap of p orbitals and is weaker because overlap is less effective.

Hybridization mixes atomic orbitals to form equivalent hybrid orbitals. sp^3 hybridization gives four tetrahedral orbitals, as in CH_4 with bond angle 109.5°. sp^2 hybridization gives three trigonal planar orbitals, as in BF_3 or ethene, with 120°

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angles. sp hybridization gives two linear orbitals, as in $BeCl_2$ or ethyne, with 180° angles.

Important Conceptual Short Questions — Topic 3

1. Why is a sigma bond stronger than a pi bond?

A sigma bond forms by direct head-on overlap, giving greater electron density between the nuclei. A pi bond forms by sideways overlap, which is less effective.

2. How does hybridization explain the tetrahedral shape of methane?

Carbon forms four equivalent sp^3 hybrid orbitals. These orbitals repel equally and point to the corners of a tetrahedron, giving bond angles of 109.5°.

3. Why does ethene have a planar structure?

Each carbon in ethene is sp^2 hybridized. The three sp^2 orbitals lie in one plane, and the unhybridized p orbitals overlap sideways to form a pi bond, locking the molecule in a plane.

4. How does sp hybridization explain the linear shape of ethyne?

Each carbon uses two sp hybrid orbitals oriented 180° apart. The two remaining p orbitals form two pi bonds, giving a linear C≡C arrangement.

Topic 4 — VSEPR model and molecular shapes

Q.4 Use VSEPR theory to predict the shapes and bond angles of molecules and ions.

VSEPR theory states that electron pairs around a central atom repel one another and arrange themselves as far apart as possible. Lone pairs repel more strongly than bond pairs, so they reduce bond angles. The shape is named from the positions of atoms, not lone pairs.

Common examples: CH_4 is tetrahedral, 109.5°; NH_3 is trigonal pyramidal, about 107°; H_2O is bent, about 104.5°; BF_3 is trigonal planar, 120°; CO_2 is linear, 180°; PCl_5 is trigonal bipyramidal; SF_6 is octahedral. VSEPR is useful in drug design because molecular shape affects how a drug fits a biological receptor.

Species	Electron pairs	Shape	Angle
CH_4	4 bond pairs	Tetrahedral	109.5°
NH_3	3 bond + 1 lone	Trigonal pyramidal	107°
H_2O	2 bond + 2 lone	Bent	104.5°
BF_3	3 bond pairs	Trigonal planar	120°
CO_2	2 bonding regions	Linear	180°

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Important Conceptual Short Questions — Topic 4

1. Why is the bond angle in water smaller than in ammonia?

Water has two lone pairs on oxygen, while ammonia has one. Lone pair-lone pair repulsion is stronger than lone pair-bond pair repulsion, so water's bond angle is compressed more.

2. How does VSEPR theory explain the linear shape of CO_2?

Carbon has two regions of electron density in the two C=O bonds. These regions repel equally and arrange 180° apart, giving a linear molecule.

3. Why does NH_3 have a pyramidal shape rather than trigonal planar?

Nitrogen has three bond pairs and one lone pair. The four electron pairs adopt a tetrahedral arrangement, but the lone pair is not seen in the molecular shape, so NH_3 is pyramidal.

4. How can molecular shape affect the action of a drug?

A drug must fit the shape and charge distribution of its receptor. VSEPR helps predict molecular geometry, so chemists can design molecules that bind more effectively.

Topic 5 — Molecular orbital theory

Q.5 Explain molecular orbital theory and use it to calculate bond order and explain the paramagnetism of oxygen.

Molecular orbital theory treats electrons as belonging to the whole molecule. Atomic orbitals combine to form bonding molecular orbitals, which are lower in energy, and antibonding molecular orbitals, which are higher in energy. Electrons fill these orbitals according to Aufbau, Pauli and Hund rules.

Bond order = ½(number of bonding electrons – number of antibonding electrons). H_2 has bond order 1, N_2 has 3, O_2 has 2, F_2 has 1 and He_2 has 0. Oxygen is paramagnetic because its MO diagram leaves two unpaired electrons in antibonding π* orbitals.

Important Conceptual Short Questions — Topic 5

1. Why does He_2 not exist according to MOT?

He_2 would have two bonding and two antibonding electrons. Its bond order is ½(2–2) = 0, so there is no net bonding.

2. How does bond order relate to bond strength and bond length?

A higher bond order means more net bonding electrons, giving a stronger and usually shorter bond. A bond order of zero means no stable bond forms.

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3. Why is oxygen paramagnetic even though its Lewis structure shows paired electrons?

The Lewis structure cannot show the correct distribution of electrons in molecular orbitals. MOT places two unpaired electrons in degenerate π* orbitals, making O_2 paramagnetic.

4. How do antibonding electrons affect molecular stability?

Antibonding electrons reduce electron density between the nuclei and oppose bonding. More antibonding electrons lower bond order and make the molecule less stable.

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